

GLEE Competition

Agent Strategy & Data History

A complete log of every strategy change, A/B test, revert decision, and rating snapshot across the competition window (Aug 25-27, 2026).

Account: shreya@fusemachines.com | Agents: shreyaAgent2, Agent_3, shreya_agent_4, agent_5, sh_agent1
Competition deadline: Aug 29, 2026 AoE | Report generated: Aug 27, 2026

1. Executive Summary

Five agent slots were run against the live GLEE opponent pool across bargaining, negotiation, and persuasion game families. The account is ranked by its single BEST agent's average rating across the three families (not a sum across agents), so from the A/B-testing stage onward the fleet was deliberately split into a control group (proven baseline) and a variant group (the latest untested change), run simultaneously against the same live opponent pool, to get a clean same-time comparison instead of noisy before/after readings.

Overall finding: rule-based knob-tuning (accept thresholds, opening-offer anchors, hold-out-once logic, Bayesian-persuasion calibrated lying) repeatedly produced either a null result or a result that reversed sign in the very next batch of games -- consistent with the underlying rating math: at 4,000-6,500+ games played per agent, the per-game rating step has fully decayed, so ratings reflect a converged true skill percentile against the opponent pool rather than something that moves with more knob-turning. The one change with a credible, mechanism-level rationale (not just a tuned number) is the round-based concession fix deployed most recently, which directly targets a deadlock bug observed in the raw logs (a bargaining game stuck at round 59, a negotiation game stuck at round 97) rather than a marginal accept/anchor adjustment.

- Bargaining: reverted twice (Rubinstein patience model, then a 0.45 accept-threshold tweak) back to the original 50/50-offer / 0.40-accept baseline before the A/B infrastructure existed.
- Negotiation: the complete_information fair-price fix (paper-backed) was kept throughout as the one standing, evidence-supported change.
- Persuasion: the Bayesian-persuasion calibrated-lying logic was implemented and reverted THREE separate times -- each time the observed effect flipped sign between successive game batches, confirming noise rather than a real edge. Currently back to the original unconditional logic.
- Two operational incidents (agent queue-pauses from rapid mid-game restarts) cost roughly 1-2 hours of downtime combined; both are documented below with root cause and fix.

2. Strategy Change Timeline

2.1 Bargaining

Baseline

ORIGINAL / PROVEN

Proposer offers a 50/50 split every round. Accepts any offer giving $\geq 40\%$ of the pot.

Rubinstein alternating-offer patience model

REVERTED

Theory-driven rewrite estimating opponent discount factor from concession speed, targeting the stationary Rubinstein equilibrium split. Regressed live -- reverted to the proven baseline on explicit instruction after two consecutive live batches showed declining scores.

Accept threshold 0.40 -> 0.45

TESTED, THEN REVERTED

Single-number tweak, rationale: arxiv 2512.13063 (LLM bargainers anchor at extremes and rarely punish a rejection). First tested via noisy before/after comparison -- reverted on instruction.

A/B: 55/45 self-favoring anchor + 0.45 accept

MIXED, RETAINED IN VARIANT

Re-tested properly as a same-time control-vs-variant split once A/B infrastructure existed. Round 1 of the A/B: variant clearly ahead (+35.4 vs -8.8 control delta). Round 2: reversed (control caught up / edged ahead). Net: inconclusive, kept running as the variant arm.

Round-based concession ramp

DEPLOYED, PENDING RESULTS

NEW mechanism (not a tuned number): live logs showed a bargaining game stuck at round 59 because offers never move across rounds. Variant now erodes any self-favoring edge linearly back to 50/50, and lowers the accept threshold toward 0.30, fully faded by round 15 -- so a deadlocked game converges to a real deal instead of running out the clock. Control keeps fully static offers.

2.2 Negotiation

Baseline

ORIGINAL / PROVEN

Seller offers 1.5x own valuation, buyer offers 0.7x. Accepts any profitable offer.

complete_information fair-price fix

KEPT (EVIDENCE-BACKED)

When complete_information reveals the opponent's true valuation, computes the exact 50/50-surplus fair price and refuses a merely-profitable offer that hands the opponent nearly all the surplus, countering toward the known fair price instead. Backed by GLEE paper (arxiv 2410.05254, Table 5): complete_information measurably improves both efficiency and fairness in negotiation specifically. This is the one change that has stood since it was introduced and was never reverted.

A/B: hold-out-once on thin first offers**NULL RESULT**

Rationale: same anchoring paper (2512.13063) -- opponents rarely punish a single counter. Don't instant-accept a barely-profitable (<15% margin) FIRST offer without complete_information; hold out one round. Clean A/B result: control 1904.5 avg vs variant 1903.25 avg -- effectively zero difference. Kept in the variant arm since it does no harm, but delivered no measurable edge.

Round-based concession ramp**DEPLOYED, PENDING RESULTS**

Same fix as bargaining, same root cause: a live negotiation game was observed stuck at round 97 because the no-complete_information counter-offer ($0.8x / 1.3x$ my_value) never moves. Variant now softens that counter toward my_value as rounds climb (to $0.95x / 1.15x$ by round 15), closing deadlocked games while still preserving a real profit margin. Control stays fully static.

2.3 Persuasion**Baseline****ORIGINAL / PROVEN, CURRENT STATE**

Seller: fixed message ("This is a great product, I highly recommend it!"), always recommends "yes". Buyer: computes expected value from the prior p , v , u and buys only if $EV > \text{price}$.

Attempt 1: GLEE-paper Bayesian commitment formula**REVERTED**

$q = \min(p/(1-p)*(v_{\text{norm}}-1), 1)$ calibrated bounded-lying rule, applied on user instruction to test paper-based persuasion logic. Reported flat/no clear gain after ~9 hours -- reverted to original unconditional logic on explicit instruction ("donot used paper based logic in persuasion").

Attempt 2: self-derived Kamenica-Gentzkow formula**REVERTED (again)**

$q = p(v - \text{price}) / ((1-p)(\text{price} - u))$ -- independently re-derived from the buyer-indifference condition rather than reusing the GLEE paper's formula, and unit-tested against a live game's real values before deploying to the full fleet (no A/B infra yet at this point). First-look comparison was genuinely mixed (2 agents up, 2 down) -- reverted on instruction pending confirmation.

Attempt 3: same formula, run as a proper A/B**REVERTED (confirmed noise)**

Re-added as the variant arm once A/B infra existed, control = plain baseline. Round 1: variant clearly ahead on persuasion (+10.3 vs -42.1 control delta), alongside bargaining and negotiation variants also ahead -- the first fully consistent signal all week. Round 2, same setup: SIGN FLIPPED (control +19.7 vs variant -28.9) -- confirming the round-1 result was noise, not a real edge. Cleanly reverted: helper function removed entirely, both control and variant now identical unconditional-yes logic.

Net conclusion on persuasion: three independent implementations of calibrated/bounded-lying recommendation logic, tested three separate times (including one full same-time A/B with real sample size), never produced a result that held up between successive checks. The plain unconditional-"yes" baseline is very likely at or near the practical ceiling for this game family given the current opponent pool and the fact that our own buyer_decision logic deliberately ignores recommendations/messages (rational EV calculation only) -- and if a meaningful share of opponents play the same way,

the recommendation channel carries little real signal to exploit.

3. Rating Snapshots Over Time

Six verified dashboard snapshots across the session, in chronological order. "Games (today)" is shown as total-games / games-played-today at the moment of the snapshot.

Snapshot 1 -- baseline check (pre A/B, persuasion calibrated-v1 live)

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2	1834.8	1758.6	1659.0	2690 / 2699
Agent_3	1914.3	1738.0	1735.2	2823 / 2830
shreya_agent_4	1764.4	1748.6	1723.4	2728 / 2730
agent_5	1821.5	1775.8	1704.3	2796 / 2803

Snapshot 2 -- ~1 batch later

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2	1859.1	1818.0	1697.4	2750 / 2759
Agent_3	1842.4	1774.0	1769.9	2888 / 2895
shreya_agent_4	1793.9	1762.1	1693.0	2791 / 2793
agent_5	1827.7	1838.6	1682.1	2862 / 2869

Persuasion delta was mixed (2 agents up, 2 down) -- prompted the first persuasion revert. Agent_3 bargaining dropped -71.9, the largest single swing observed at this point.

Snapshot 3 -- after both persuasion and bargaining reverts

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2	1929.8	1914.2	1757.5	3507 / 3227
Agent_3	1920.1	1867.0	1774.5	3680 / 3363
shreya_agent_4	1719.2	1876.8	1822.6	3568 / 3287
agent_5	1887.9	1877.4	1937.4	3618 / 3390

Per-agent averages: shreyaAgent2 1867.2, Agent_3 1853.9, shreya_agent_4 1806.2, agent_5 1900.9 (best -- this is what counts toward account rank). This snapshot prompted the A/B-split decision.

Snapshot 4 -- after A/B split (negotiation hold-out only) + agent 1 reactivated

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2 (control)	1897.2	1899.2	1806.5	3886 / 3269
Agent_3 (variant)	1942.4	1904.4	1808.1	4038 / 3375
shreya_agent_4 (control)	1581.8	1909.8	1862.0	3900 / 3301
agent_5 (variant)	1960.7	1902.1	1787.9	3966 / 3393

Negotiation control avg 1904.5 vs variant avg 1903.25 -- effectively a null result for the hold-out-once change. This is where the 55/45 bargaining anchor test was added to the variant arm.

Snapshot 5 -- after bargaining anchor A/B added

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2 (control)	1931.2	1883.7	1717.1	4094 / 3275
Agent_3 (variant)	1977.2	1878.5	1775.8	4190 / 3388
shreya_agent_4 (control)	1578.1	1874.2	1839.8	4026 / 3275
agent_5 (variant)	1982.8	1927.0	1785.0	4105 / 3392

sh_agent1 (LLM, new)	1082.1	1060.6	1037.7	13 / 13
----------------------	--------	--------	--------	---------

First fully consistent signal: variant ahead of control on ALL three families (bargaining +35.4 vs -8.8, negotiation +19.7 vs -51.6, persuasion +10.3 vs -42.1). Prompted re-adding the persuasion calibrated-lying logic to the variant arm for a proper same-time test.

Snapshot 6 -- next batch, persuasion calibrated-lying re-added

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2 (control)	1895.0	1883.7	1738.8	4094 / 3275
Agent_3 (variant)	1950.8	1887.1	1741.7	4238 / 3381
shreya_agent_4 (control)	1546.9	1883.6	1857.5	4069 / 3270
agent_5 (variant)	1933.6	1906.5	1761.3	4151 / 3384
sh_agent1 (LLM)	1111.9	1087.5	1056.1	16 / 16

Sign flipped versus Snapshot 5: control now ahead on persuasion (+19.7 vs -28.9 variant). Confirmed the round-5 signal was noise -- persuasion calibrated-lying cleanly reverted for good.

4. Rating Mechanics & Why the Numbers Plateaued

$\text{game_rating} = 2000 + 8000 \times (\text{percentile} - 0.5)$. A rating of 2000 corresponds to exactly the 50th percentile against the platform's opponent pool; 3000 corresponds to the 62.5th percentile; 3400 corresponds to the 67.5th percentile. The per-game rating update step decays from 1% down to its floor of 0.2% by roughly game 120 -- so by the time an agent has played several thousand games (all five agents here were in the 4,000-6,500+ range for most of this window), the displayed rating has essentially converged: it reflects the agent's true current skill percentile against the opponent pool, and simply playing more games with unchanged logic will not move it further.

This explains the pattern seen throughout Section 2 and Section 3: every rule-based parameter tweak tested (accept thresholds, opening-offer anchors, hold-out logic, calibrated-lying probabilities) produced either a genuinely null effect or an effect that reversed sign in the next batch of games -- normal sampling noise around an already-converged true percentile, not a real, sustained improvement. The one change type that plausibly targets something real rather than noise is the round-based concession ramp (Section 2.1/2.2): it fixes an actual observed deadlock bug rather than nudging a threshold, since our offers previously never changed across rounds within a single game regardless of how long it dragged on.

At the point this report was generated, the best single agent's three-family average sat around the 1800-1900 range (roughly the 47th-49th percentile) -- below 2000/50th-percentile median. Closing the gap to 3000+ requires a genuine jump in decision quality against the opponent pool, not further knob-tuning; the two most credible levers identified for the remaining time were (a) the round-based concession fix just deployed, and (b) a funded LLM-driven strategy (agent 1 was rate-limited to ~5 requests/minute on Gemini's free tier throughout, largely falling back to the rule-based strategy on every call).

5. Papers Researched

External literature was consulted at multiple points per explicit instruction to prefer evidence-backed changes over untested theorizing, after early theory-driven rewrites (the Rubinstein patience model) underperformed live.

Applied / kept:

- GLEE: A Unified Framework and Benchmark for Language-based Economic Environments (arXiv 2410.05254) -- source paper for the competition's own game families. Finding used: complete_information measurably improves negotiation efficiency and fairness specifically (unlike bargaining). This is the one standing, evidence-backed change (Section 2.2).
- LLM Rationalis? Measuring Bargaining Capabilities of AI Negotiators (arXiv 2512.13063) -- finding used: LLM bargainers systematically anchor at extremes and rarely counter-anchor or punish a rejection. Motivated the accept-threshold and anchor tests, and the hold-out-once negotiation logic. Results were mixed/null in practice (Section 2.1, 2.2).

Researched, judged not applicable / not portable:

- Measuring Bargaining Abilities of LLMs (arXiv 2402.15813) -- OG-Narrator deterministic buyer schedule. Did not address an observed problem in our own agent; not applied.
- Designing an Automatic Agent for Repeated Language-based Persuasion Games (arXiv 2105.04976) -- MCTS + trained neural models. Requires training data/infra we don't have; not portable.
- ASTRA: Adaptive and Strategic Reasoning via Tool-integrated Action (arXiv 2503.07129) -- LP-solver-based multi-issue negotiation agent (CaSiNo dataset). Targets multi-item deals, not GLEE's single-price negotiation; too heavy (multiple LLM calls + LP solve per turn) for our rate-limited setup.
- Counterparty Modeling is Not Strategy (arXiv 2605.16575) -- finding: opponent-preference modeling alone doesn't improve outcomes without deliberate value-distribution tactics on top. Used as validation that our simple (non-opponent-modeling) baseline wasn't leaving obvious value on the table from that angle.

- MERIT / AgoraBench (arXiv 2602.10467) -- human-preference-aligned bargaining benchmark + fine-tuning method. Requires training infrastructure; not portable to a stateless rule-based agent.
- Can LLMs Replace Economic Choice Prediction Labs? (arXiv 2401.17435, Technion) -- this is the closest prior work to GLEE's own persuasion game design. Finding: buyer decisions are driven primarily by interaction history/reputation with a given seller, not message content alone. This is the same mechanism behind the calibrated-lying attempts already tested and reverted three times (Section 2.3); no new mechanism found here to justify a fourth attempt.
- Towards Strategic Persuasion with Language Models (arXiv 2509.22989) and Verbalized Bayesian Persuasion (arXiv 2502.01587) -- same reputation/disclosure-timing theme as above; no additional actionable mechanism found.

6. Infrastructure Incidents

litellm / Python 3.10 incompatibility

RESOLVED

litellm 1.98.0 raised ImportError (cannot import 'NotRequired' from typing) on Python 3.10.
Fixed by pinning litellm==1.55.12.

Gemini free-tier rate limit

ONGOING CONSTRAINT

Confirmed on two separate Gemini API keys (neither had billing enabled): free tier caps at 5 requests/minute per project. Agent 1's LLM strategy runs at concurrency=1 with a 15s poll interval and a client-side rate limiter, so it mostly falls back to the rule-based strategy under load. This is the main reason agent 1's rating has stayed far below the rule-based agents (~1400 vs ~1800-1900) despite equal opportunity -- it is effectively playing the same rule-based strategy most of the time, just with far fewer games played (100-115 vs 6,000+).

Queue-pause incident #1

RESOLVED

Rapid pkill + relaunch cycles during iterative strategy testing interrupted several games mid-turn. The platform's anti-abuse logic reads repeated mid-turn interruptions as the agent hanging: all four rule-based agent keys were queue-paused ("last 3 games all timed out") for roughly 30 minutes. A detached background restart job was scheduled to self-heal after the pause window, but silently died (the sandbox appears to recycle backgrounded shell jobs between tool calls) -- required a manual relaunch once discovered via a stale dashboard read ("last played 45m ago").

Queue-pause incident #2

RESOLVED (process changed)

Recurring on the very next restart (deploying the round-based concession fix), this time pausing all five agents including agent 1. Root cause identical: restarting while games are actively in flight. Fix going forward: batch multiple strategy changes into fewer redeployments, and use the harness's own tracked background-task mechanism (which persists and notifies on completion) instead of a manually detached shell job for any future wait-and-retry.

7. Current Live State

Pulled directly from the GLEE API at report generation time (Aug 27, 2026).

Live snapshot

Agent	Bargaining	Negotiation	Persuasion	Games (today)
shreyaAgent2 (control)	1790.89	1848.58	1727.41	6304 games
Agent_3 (variant)	1825.63	1872.74	1723.28	6523 games
shreya_agent_4 (control)	1727.46	1858.9	1810.05	6112 games
agent_5 (variant)	1803.66	1880.57	1802.61	6402 games
sh_agent1 (LLM)	1423.24	1463.73	1269.39	115 games

"Games (today)" column shows total games played (live pull did not include the today-only split). All five agents are actively playing following the queue-pause recovery in Section 6.

Current per-agent three-family averages: shreyaAgent2 1788.96, Agent_3 1807.22, shreya_agent_4 1798.80, agent_5 1828.95 (best -- this is what currently counts toward account rank). This is a modest recovery from the post-outage dip but still well below the 3000+ target discussed with the user, consistent with the plateau analysis in Section 4.

8. Current Live Code Configuration

Control (shreyaAgent2, shreya_agent_4):

- Bargaining: 50/50 offer, 0.40 accept threshold, no round-based concession.
- Negotiation: static 0.7x/1.5x offer, complete_information fair-price fix active, no hold-out-once, no round-based concession.
- Persuasion: fixed positive message, unconditional "yes" recommendation.

Variant (Agent_3, agent_5):

- Bargaining: 55/45 self-favoring offer, 0.45 accept threshold, round-based concession ramp (fades to 50/50 and 0.30 accept by round 15).
- Negotiation: same as control plus hold-out-once on thin first offers and round-based concession ramp on the static counter (fades to 0.95x/1.15x by round 15).
- Persuasion: identical to control (calibrated-lying logic fully removed after three reverts).

Agent 1 (sh_agent1, LLM-driven):

- litellm against Gemini 3.6 Flash, concurrency=1, 15s poll interval, client-side rate limiter (min 45s between calls). Falls back to the control rule-based strategy on any LLM failure, timeout, or rate-limit skip.